

Aurora PostgreSQL vs RDS PostgreSQL

Architecture (basic difference)

RDS PostgreSQL = normal PostgreSQL, oka instance ki storage (EBS) attach chesi untundi. Simple, classic setup.

Aurora PostgreSQL = AWS specially build chesina distributed system. Storage and compute separate ga untayi, storage 3 Availability Zones lo 6 copies replicate avutundi automatic ga. So idi more resilient.

Scaling differences

RDS lo storage meeru manually decide chesi provision cheyyali (max 64 TiB varaku). Aithe capacity full ayithe meere penchali.

Aurora lo storage automatic ga grow avutundi (10GB chunks lo), max 128 TiB varaku - meeru edi cheyyakkarledu, automatic.

Read replicas (traffic ekkuva ayithe add chese extra copies) - RDS lo prathi replica ki separate storage kavali (so cost penchutundi). Aurora lo anni replicas oke storage share chesukuntayi - so replicas add chesina storage cost penchadu, that's a big advantage.

Aurora Serverless v2 ane option kuda undi - traffic batti compute automatic ga penchutundi/thaggutundi. RDS lo idi ledu, meeru fixed instance run cheyyali eppudu.

Performance

Chinna workloads ki RDS bavuntundi (simple, direct).

Traffic/concurrency ekkuva ayithe (peddha apps), Aurora better perform chestundi - 2-5x throughput ekkuva ivvachu, especially write-heavy cases lo.

High Availability (failover time - DB fail ayithe backup ki switch avvadam)

RDS: 60-120 seconds padutundi switch avvadaniki.

Aurora: 30 seconds kanna thakkuva (chala fast) - kaarnam, storage already shared ga undi so promote cheyyadam fast.

Cost (pricing)

RDS: instance + provisioned storage (meeru entha tesukunte antha) + IOPS ki charge chestaru. Fixed model, predictable.

Aurora: instance rate konchem ekkuva (~20% RDS kanna) kani storage meeru actual ga use chesindi matrame charge chestaru, and replicas storage cost penchavu. So peddha scale lo (multiple replicas unte) Aurora total cost thakkuva padochu.

Bottom line - eppudu edi vaadali

RDS → Chinna/medium app, budget matter chestundi, simple setup kavalante.

Aurora → Peddha traffic, chala read replicas (3+) kavalante, fast failover kavalante, ya traffic unpredictable ga spike avutundhi ante.

Note: Exact prices AWS region batti maripothayi. Current numbers kosam AWS Pricing Calculator use chesi telusukovacchu.